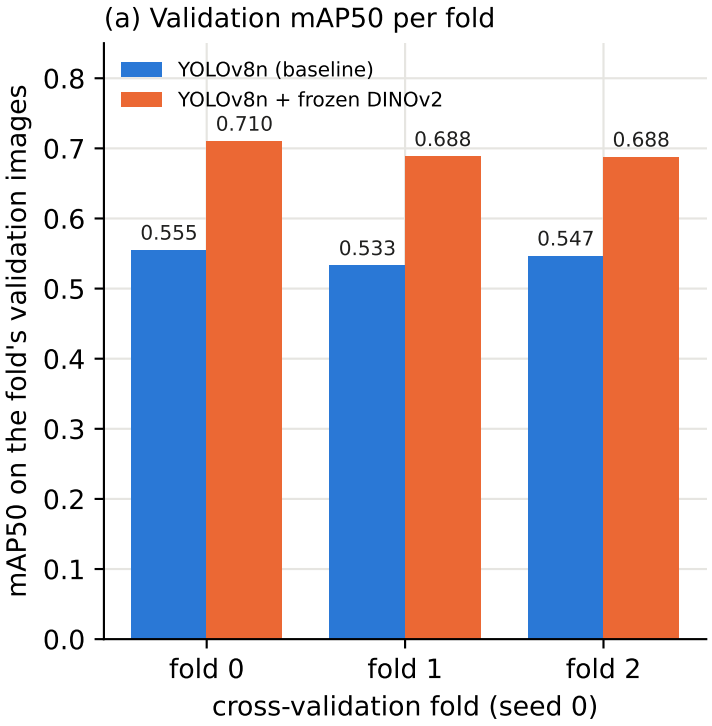
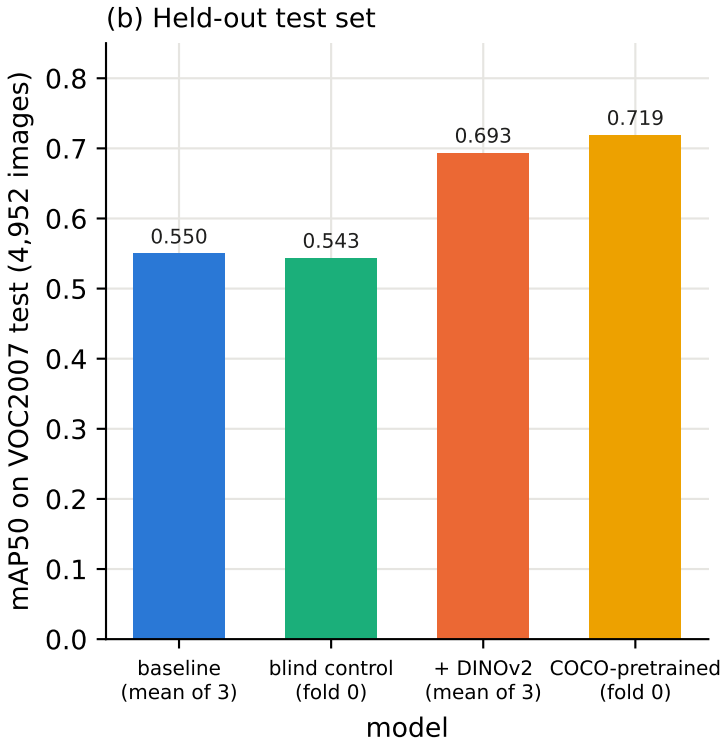
YOLOv8n with a frozen DINOv2 in front: results on PASCAL VOC2007

20 classes. 3-fold cross-validation on 5,011 trainval images; 4,952 held-out test images. Both models trained from scratch, 100 epochs, identical settings.

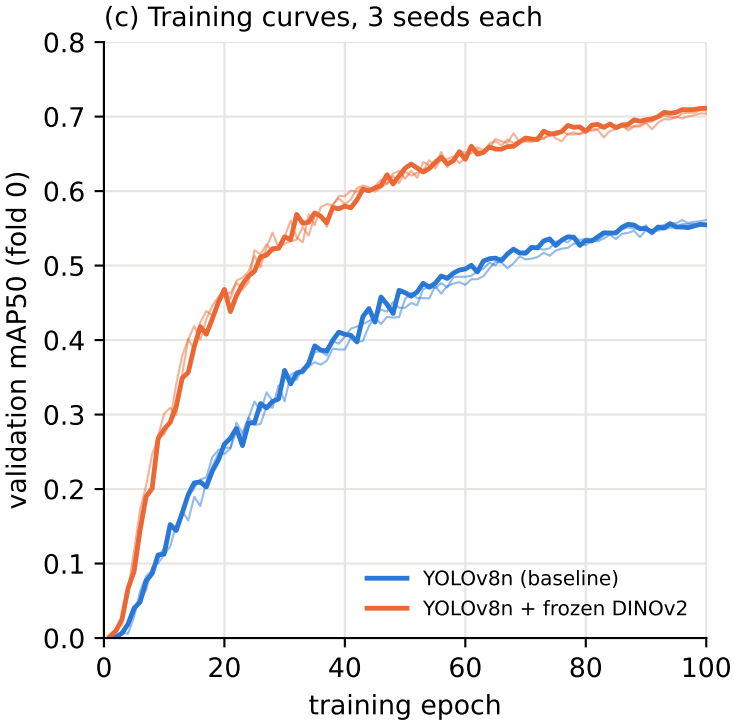
Verdict (rule fixed before training): DINOv2 helps. Mean fold gain +0.151 mAP50, 27x the seed noise (0.0056), positive on all 3 folds.



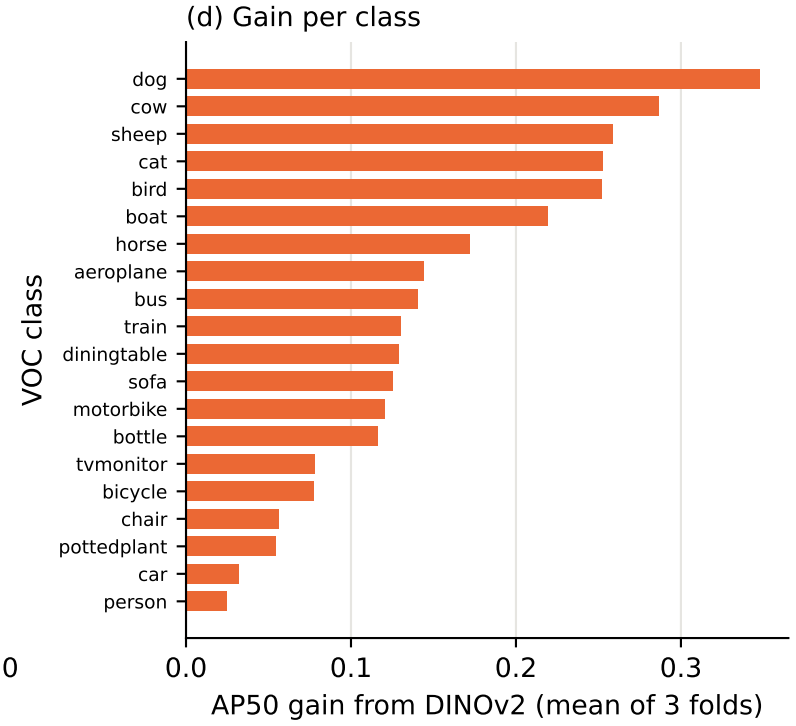
Averaged over the 3 cross-validation folds, mAP50 rose from 0.545 (YOLOv8n) to 0.696 with a frozen, pre-trained DINOv2 in front of it.



Held-out test: 0.550 -> 0.693. The blind control (same architecture, DINOv2's information removed) gained essentially nothing (0.543 vs 0.545), so the gain came from DINOv2's visual knowledge, not from the extra layers.



The YOLOv8n part of both models was trained from scratch (both start near zero) with identical settings; only DINOv2 differs between them.

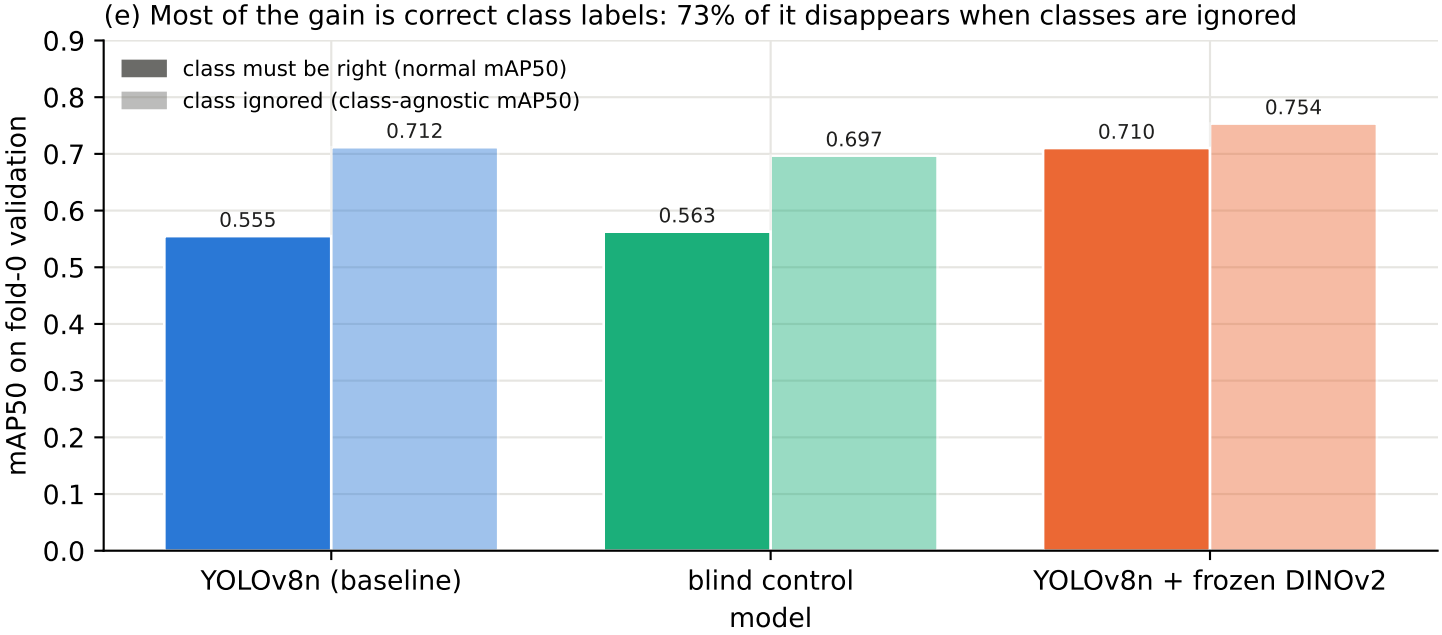


All 20 classes improve. The largest gains are animals that look alike (dog, cow, sheep, cat, bird).

Why DINOv2 helps, and what it costs

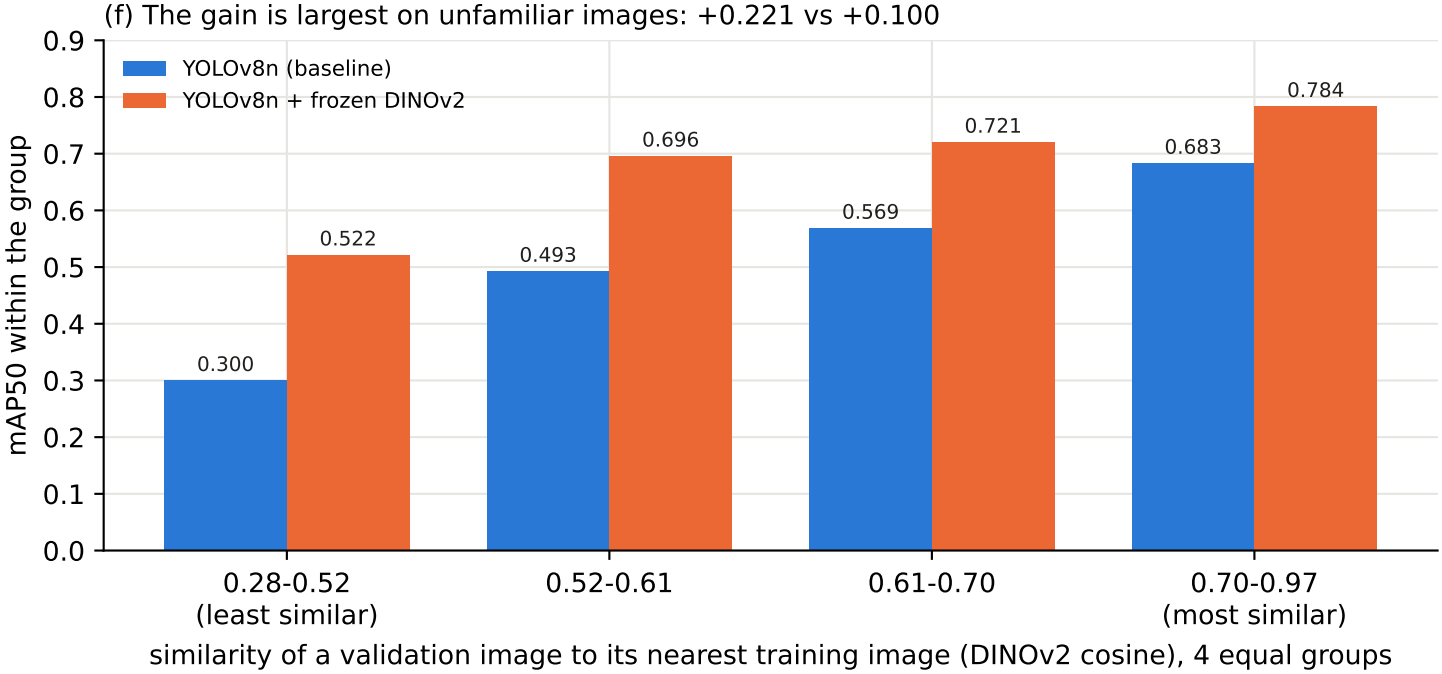
Fold 0, seed 0 models, scored on fold 0's 1,670 validation images unless stated.

(e) Most of the gain is correct class labels: 73% of it disappears when classes are ignored



About 73% of DINOv2's gain came from assigning the correct class to detected objects: the plain YOLOv8n finds most objects but often names them wrongly (for example a pony as 'dog'), and DINOv2's scene summary fixes that.

(f) The gain is largest on unfamiliar images: +0.221 vs +0.100



(g) Cost: forward pass, fp32, 640x640, RTX 4070 laptop GPU. DINOv2's 22M weights are frozen.

model	parameters (M)	trainable (M)	GFLOPs at 640 px	ms/image, batch 1	ms/image, batch 16
YOLOv8n	3.01	3.01	8.1	6.3	1.9
YOLOv8n + DINOv2	25.27	3.21	19.3	11.8	3.8

Example predictions on fold-0 validation images

Left: baseline. Right: + DINOv2. Confidence threshold 0.25. The baseline calls a pony and a hen 'dog' and a kitten 'person'.

